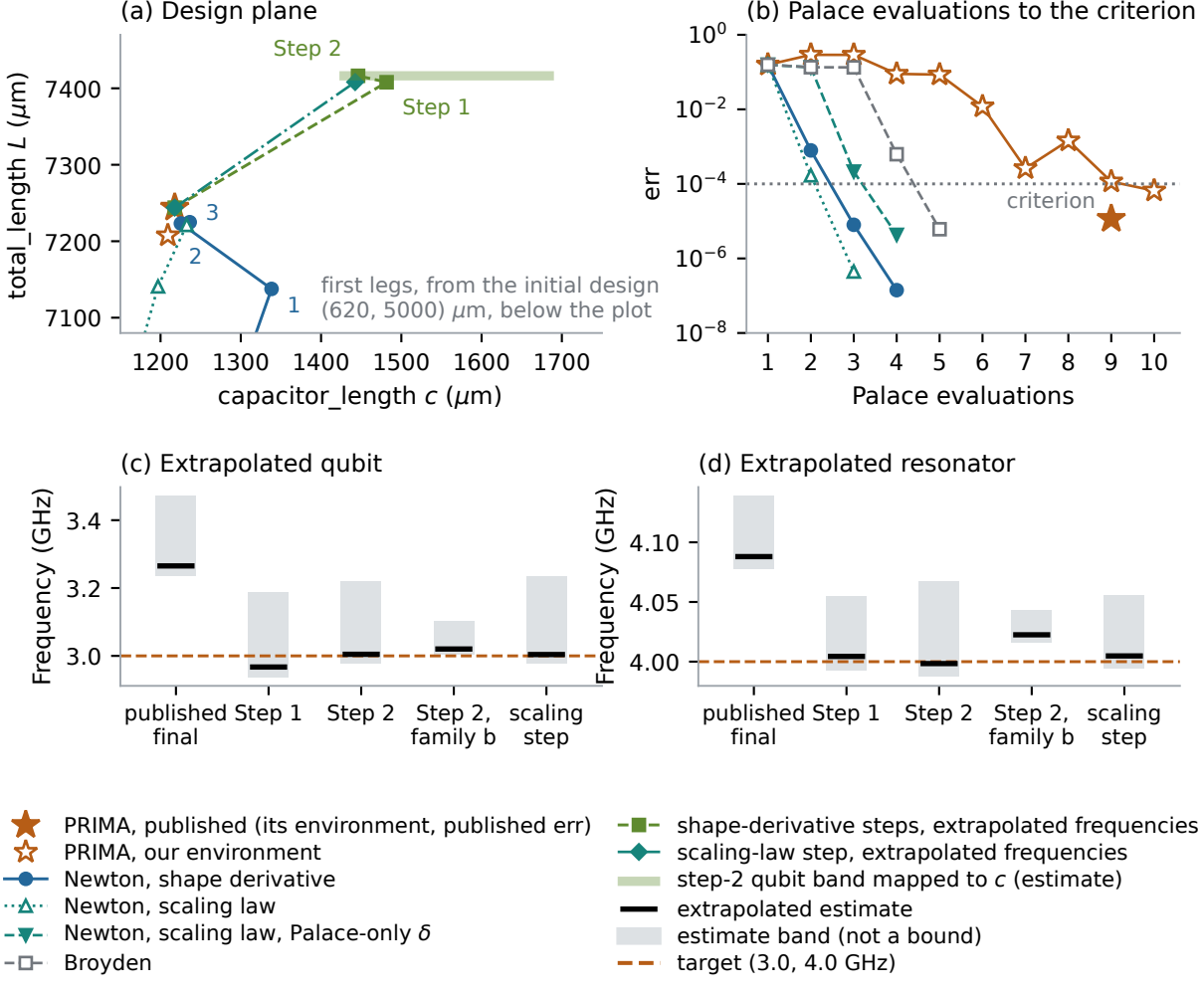
(a) Design plane
Step 2
Step 1
3
2
1
first legs, from the initial design
(620, 5000) μm, below the plot
total_length L (μm)
7400
7300
7200
7100
1200
1300
1400
1500
1600
1700
capacitor_length c (μm)
(b) Palace evaluations to the criterion
err
criterion
Palace evaluations
(c) Extrapolated qubit
Frequency (GHz)
3.4
3.2
3.0
published final
Step 1
Step 2
Step 2, family b
scaling step
(d) Extrapolated resonator
Frequency (GHz)
4.10
4.05
4.00
published final
Step 1
Step 2
Step 2, family b
scaling step
PRIMA, published (its environment, published err)
PRIMA, our environment
Newton, shape derivative
Newton, scaling law
Newton, scaling law, Palace-only δ
Broyden
shape-derivative steps, extrapolated frequencies
scaling-law step, extrapolated frequencies
step-2 qubit band mapped to c (estimate)
extrapolated estimate
estimate band (not a bound)
target (3.0, 4.0 GHz)